

NeuroSpark — Study Plan Guide

What to learn (and in what order) to actually understand and build Stage 1.5

Study while you build. Learn each topic just before the build phase that needs it — not all up front.

This guide turns the go-forward plan into a learning roadmap: seven knowledge tracks, each tied to a build phase, with concrete resources and a 'done when' checkpoint. It assumes you already have working Python + NumPy and the running Stage 1 codebase.

Companion documents in this folder: the verdict/plan (NeuroSpark_Stage1_5_Viability_and_GoForward_Plan.pdf) and the evidence (NeuroSpark_DeepResearch_Findings.pdf). Generated 22 June 2026.

0. How To Use This Guide

- **Just-in-time, not up-front.** The fastest way to stall is to try to master Friston's free energy before writing any code. Instead, learn each track to the depth the next build phase needs, build that phase, then move on.
- **Build the toy first, read the theory second.** For every concept, implement a 30-line version yourself before reading the heavy paper — understanding follows from the code.
- **One notebook per track.** Keep a Jupyter notebook of tiny experiments per track; these become your figures and your intuition.
- **You don't need all of it.** See §4 'Minimum Viable Path' if time is short.

1. Where You Already Are

Skill	Status	Action
Python, NumPy, SciPy	Have it (project runs)	None — you're fine
Brian2 / spiking networks	Used it, shallow understanding	Track A — deepen the 'why'
Active inference / free energy	MSE proxy only (not real yet)	Track B — the big one
Hebbian plasticity	Coded a version	Track C — light top-up
Graph theory / connectomes	Synthetic graph only	Track D — for the real FlyWire data
Reinforcement learning	None in repo	Track E — needed for the baseline
Statistics / experiment design	Minimal	Track F — needed to publish

2. The Seven Knowledge Tracks

Priority key: **CORE** = the claim depends on it; **SUPPORT** = needed but lighter; **POLISH** = for the writeup. Times are part-time estimates.

Track A — Spiking Neural Networks & Brian2

SUPPORT · ~1 week

Why it matters here: Your substrate is 1,000 LIF neurons in Brian2. You must understand membrane dynamics, thresholds, refractoriness, and why spike timing (not just rate) can carry information — the 'rate-coded vs spiking' ablation depends on it.

Learn:

- LIF neuron model: membrane equation, $V_{\text{rest}}/V_{\text{th}}/\text{reset}$, refractory period, time constant τ
- Synaptic transmission, excitatory vs inhibitory currents, Poisson background drive
- Rate coding vs temporal coding; population activity and rasters
- Brian2 specifics: NeuronGroup, Synapses, Network store/restore, monitors, run()

Best resources:

- Gerstner, Kistler, Naud, Paninski — *Neuronal Dynamics* (free online book + EPFL lectures), chapters 1–4 — the standard gentle intro.
- Brian2 official tutorials (brian2.readthedocs.io) — do all three notebooks.
- Your own `brain/network.py` + `test_brian.py` — annotate every line.

Done when: You can explain, without notes, why your motor layer can go silent and how `S_direct` fixes it.

Track B — Active Inference & the Free Energy Principle

CORE · ~3–4 weeks

Why it matters here: This IS your novelty. Right now the code uses MSE as a stand-in; until you implement real variational free energy (VFE) for perception and expected free energy (EFE = risk + ambiguity – epistemic value) for action, the 'active inference' claim is not earned and reviewers will reject it.

Learn:

- Bayesian inference, generative models, the variational/ELBO trick (complexity – accuracy)
- Discrete active inference: the A (likelihood), B (transitions), C (preferences), D (prior) matrices
- VFE for perception (state inference) vs EFE for policy selection
- Epistemic value (information gain) vs pragmatic value (goal-seeking) — the curiosity term
- Precision / temperature and softmax action selection

Best resources:

- Bogacz (2017), 'A tutorial on the free-energy framework for modelling perception and learning', J. Math. Psych. — the clearest on-ramp; work the exercises.
- Smith, Friston, Whyte (2022), 'A step-by-step tutorial on active inference and its application to empirical data' — canonical, with code.
- **pymdp** (Heins et al., 2022) — Python library for discrete active inference; replicate its grid-world agent, then graft your connectome decoder onto it.
- Parr, Pezzulo & Friston (2022), *Active Inference* (MIT Press, free PDF) — reference, not a first read.
- Sajid et al. (2021) 'Active Inference: Demystified and Compared' — frames AIF vs RL (your paper's core).

Done when: You can derive EFE for one action by hand AND your code's free energy provably drops over an episode.

Track C — Synaptic Plasticity (Hebbian / STDP)

SUPPORT · ~3–4 days

Why it matters here: Your learning is local plasticity, not backprop. You need to defend why FEP-modulated Hebbian updates are legitimate and how they relate to STDP (which matters for the Stage 2 story).

Learn:

- Hebbian rule and its instabilities; weight normalization / clipping
- Spike-timing-dependent plasticity (STDP) windows
- Three-factor / neuromodulated plasticity (your 'free-energy-modulated' rule is one)

Best resources:

- Neuronal Dynamics, plasticity chapters (Gerstner et al.).
- Frémaux & Gerstner (2016), 'Neuromodulated STDP and the three-factor learning rule'.
- Your bridge/learning_rules.py — map each term to the theory.

Done when: You can justify your update rule as a three-factor learning rule in one paragraph.

Track D — Connectomics, Graph Theory & FlyWire

CORE · ~1–1.5 weeks

Why it matters here: Phase 1 swaps the synthetic graph for a real FlyWire subgraph. You need graph metrics to run the topology ablation and to defend that your <1% subgraph is functionally meaningful.

Learn:

- Directed weighted graphs; degree distributions; small-worldness; rich-club; path length
- Watts–Strogatz vs random vs biological topology — what each ablation actually tests
- FlyWire data model: directed synapse-count weights, predicted neurotransmitter signs, thresholds
- Selecting a coherent sensorimotor subgraph by cell type (avoid arbitrary cuts)

Best resources:

- FlyWire: Dorkenwald et al. & Schlegel et al. (Nature 2024) — the connectome + annotations papers.
- Tools: **navis** / **fafbseg-py**, **CAVEclient**, Codex (codex.flywire.ai); NetworkX for metrics. Data: Zenodo v783 (DOI 10.5281/zenodo.10676866).
- NetworkX tutorial + a short read on small-world / rich-club coefficients.

Done when: You can extract a labelled ~1k-neuron subgraph and report its degree/rich-club stats vs a WS graph.

Track E — Reinforcement Learning (the baseline)

CORE · ~1.5–2 weeks

Why it matters here: Your headline result is 'Active Inference vs RL'. You must implement a fair RL baseline (Q-learning and/or PPO) on the same environment, encoder, and decoder — a weak baseline invalidates the whole comparison.

Learn:

- MDPs, value functions, Q-learning, exploration (ϵ -greedy) — the tabular foundation
- Policy gradients and PPO (actor–critic, clipped objective)
- Sample efficiency & learning curves — the metric you compare on
- Fair-comparison hygiene: same obs/reward/seeds/budget for AIF and RL

Best resources:

- Sutton & Barto, *Reinforcement Learning: An Introduction* (2nd ed., free) — chs 3–6 (MDPs, TD, Q-learning) are enough for the tabular baseline.
- OpenAI **Spinning Up in Deep RL** — the PPO page + reference implementation.
- **CleanRL** ppo.py — a single-file, readable PPO to adapt.

Done when: Q-learning and PPO both train on your grid with the same interface and produce learning curves.

Track F — Experimental Design & Statistics

CORE · ~4–5 days

Why it matters here: Phase 3–4 is 50 trials × 6 conditions. Without correct statistics the paper is unpublishable; with them, this is the part reviewers respect most.

Learn:

- Trials vs episodes; seeds; what to randomize and hold fixed; pre-registering the ablations
- Mean ± SEM, confidence intervals, t-tests, Mann–Whitney (when not normal)
- Effect size (Cohen's d) and why p-values alone aren't enough
- Multiple-comparison correction (you compare many conditions)

Best resources:

- [scipy.stats docs \(ttest_ind, mannwhitneyu\)](#) — practical.
- Any short 'statistics for experimenters / for ML' primer; Cohen's d explainer.
- Reproducibility: fix seeds, save a single results file, one-command rerun.

Done when: Your ablation bar chart shows correct error bars, p-values, and effect sizes from a reproducible run.

Track G — Scientific Writing & Reproducibility

POLISH · ~1 week (overlaps writing)

Why it matters here: A correct study still fails if the prior-art section is dishonest or the repo won't run. This track is what converts results into an accepted workshop paper.

Learn:

- Paper structure (intro / methods / results / discussion); honest related-work positioning
- Making figures that stand alone; writing a tight abstract
- Repo hygiene: README 5-minute demo, pinned requirements, one-command reproduce
- Preprinting (arXiv/bioRxiv) and choosing a workshop (IWA, NeuroAI)

Best resources:

- Read 2–3 accepted IWA / NeuroAI workshop papers to calibrate scope and length.
- Whitesides, 'Writing a paper' (short, classic) for structure.
- Your own go-forward plan §3.4 lists the exact venues and the bar.

Done when: A stranger can clone the repo, run one command, and regenerate your main figure.

3. A 12-Week Study-While-You-Build Schedule (part-time)

Tracks are interleaved with the build phases from the go-forward plan, so you always learn right before you apply. Adjust to your pace.

Weeks	Study (just-in-time)	Build milestone
1	Track A (SNN/Brian2) + Phase 0 hygiene	Repo cleaned, FE logging fixed, .gitignore
2	Track D (connectomics/FlyWire)	Phase 1: real labelled subgraph in Brian2; raster looks structured
3–5	Track B (active inference) — the core block	Phase 2: true VFE/EFE; free energy provably decreases
6	Track C (plasticity) top-up	Plasticity rule documented; P0 bugs retired
7–8	Track E (RL baseline)	Phase 3a: Q-learning + PPO run on same interface
9–10	Track F (stats) + finish ablations	Phase 3b: 50 trials × 6 conditions; results file
11–12	Track G (writing)	Phase 4: 5 figures + 6-page draft; preprint

4. Minimum Viable Path (if time is short)

If you can only do part of this, do exactly these — they are the load-bearing pieces of the publishable claim:

- **Track B (active inference)** to the depth of implementing real EFE with an epistemic term — non-negotiable; without it there is no 'active inference' paper.
- **Track D** just enough to load a real, cell-type-coherent FlyWire subgraph.
- **Track E** a solid *tabular Q-learning* baseline (skip PPO if needed).
- **Track F** the four basics: trials/seeds, mean±SEM, a t-test, Cohen's d.

Skip for now: heavy STDP theory, deep PPO internals, and all of Stage 2 wetware.

5. Resource Quick-List

Topic	Go-to resource
Spiking neurons	Gerstner et al., <i>Neuronal Dynamics</i> (free book + EPFL videos)
Brian2	brian2.readthedocs.io tutorials
Active inference (intro)	Bogacz 2017 tutorial (J. Math. Psych.)
Active inference (canonical)	Smith, Friston & Whyte 2022 step-by-step tutorial
Active inference (code)	pymdp library (Heins et al. 2022)
AIF vs RL framing	Sajid et al. 2021, 'Demystified and Compared'
Plasticity	Frémaux & Gerstner 2016 (three-factor rule)
FlyWire data	Dorkenwald/Schlegel Nature 2024; Codex; navis/CAVEclient; Zenodo v783
RL textbook	Sutton & Barto, 2nd ed. (free PDF)
PPO implementation	OpenAI Spinning Up; CleanRL ppo.py
Statistics	scipy.stats; a short effect-size / multiple-comparisons primer
Venues	IWAI (iwaiworkshop.github.io); NeuroAI workshop

Bottom line: the single highest-leverage thing you can study is **Track B (real active inference)** — it is both the hardest topic and the one your entire contribution rests on. Start there once Phase 0–1 give you a real connectome

to run it on.

Prepared for Project NeuroSpark · Study Plan Guide · 22 June 2026. Resource titles are given by name so they remain findable even as URLs change.